THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

[MSMK7026] [Data and AI Transformation]

GENERAL INFORMATION

Instructor: Mr. Kevin Cheung

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Consultation times: By appointment

Tutor: TBD

Pre-requisites: [ ]
Co-requisites: [ ]
Mutually exclusive: [ ]

Course Website: TBD
Reference Book:
[Book]
The following programming languages are used in this course: [programing language]
[Software] is also used.

COURSE DESCRIPTION

In an era of accelerating technological disruption, Data and Artificial Intelligence (AI) are reshaping how organizations compete, operate, and create value. While early waves of transformation focused on traditional analytics—such as descriptive reporting, predictive modeling, and machine learning—recent advancements in Generative AI and emerging agentic AI systems are redefining the boundaries of automation, decision-making, and human–machine collaboration.

This course examines Data & AI Transformation from a strategic, organizational, and leadership perspective. It explores how firms move beyond isolated AI use cases to build enterprise-wide capabilities that integrate traditional analytics, Generative AI, and agentic AI. Rather than focusing solely on technical implementation, the course emphasizes how organizations align business strategy, governance, talent, risk management, and technology architecture to unlock sustainable value from AI investments. Students will critically evaluate both the opportunities and limitations of AI technologies—including issues of scalability, data governance, hallucination risk, and operational integration.

Drawing on real-world case studies across industries, students will analyze transformation through four core dimensions:

- WHY: Strategic ambition and value creation
- WHAT: Use case ideation, framing and prioritization
- WHO: Organizational structure, talent, and opportunities
- HOW: Data ecosystems, AI platforms and governance

Through lectures, structured frameworks, applied workshops, and project-based learning, students will develop the analytical rigor, problem-structuring capability, and executive communication skills required to lead Data & AI transformation in complex global organizations.

COURSE OBJECTIVES

Structured Framework for Problem-Solving: Through a series of real-life case studies spanning various industries, students will gain a structured framework to approach data and AI transformation challenges. By analyzing these cases, students will develop a deep understanding of the underlying principles and methodologies essential for effective problem-solving in complex business environments.

Application-Oriented Learning: The course goes beyond theoretical concepts by providing ample opportunities for students to apply their knowledge in practical settings. Practical skills training, interactive assignments, and guest speakers from diverse backgrounds, will offer invaluable insights into real-world applications of data analytics.

PROGRAMME LEARNING OUTCOMES

PLO1: Acquisition and internalization of knowledge of the programme discipline

PLO2: Application and integration of knowledge

PLO3: Inculcating professionalism and leadership

PLO4: Developing Global Outlook

PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1 – Conceptual Mastery Explain the evolution of enterprise AI from traditional analytics (descriptive, predictive, prescriptive) to Generative AI and emerging agentic AI systems, and assess their implications for business transformation.	PLO [1&4]
CLO2 – Structured Transformation Thinking Apply the WHY–WHAT–WHO–HOW transformation framework to diagnose an organization's AI maturity across vision, use case pipeline, operating model, and technology platforms.	PLO [2]
CLO3 – Use Case & Value Prioritization Ideate, frame, evaluate and prioritize AI use cases across industries using structured pipeline management principles (from ideation to pilot to scale), considering feasibility, value creation, and organizational readiness.	PLO [2]
CLO4 – Governance Critically assess governance, risk, regulatory, and other considerations associated with deploying predictive, generative, and agentic AI systems in global business environments.	PLO [1&4]
CLO5 – Executive Communication & Leadership Develop and deliver professional executive-level presentations articulating AI transformation strategies, demonstrating structured problem-solving and persuasive communication.	PLO [3&5]

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected Hour	Study Load (% of study)
T&L1. Interactive lectures and workshops	30	25%
T&L2. Exercises & Assignments	60	50%
T&L3. Lecture Preparation & Self-study	30	25%
Total	120	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes

A1. In-class participation	Attendance & discussions	30%	CLO 1, 2, 3, 4 & 5
A2. Individual Assignments	Effort & quality	30%	CLO 1, 3 & 5
A3. Group Project	Effort & quality	40%	CLO 2, 4 & 5
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	•Demonstrate a strong understanding of all relevant knowledge •Handling questions professionally •High participation in discussions •Present arguments that have an element of originality •Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills •Excellent performance in assignments
B+, B, B-	•Demonstrate a good understanding of all relevant knowledge •Handling questions in a logical way •Good participation in discussions •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills •Good performance in assignments
C+, C, C-	•Demonstrate a basic understanding of the concepts involved •Fairly address questions as set •Some participation in discussions •Present arguments in a well-structure manner •Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills •Acceptable performance in assignments
D+, D	•Demonstrate a minimum understanding of the concepts involved •Barely address questions as set •Minimal or no participation in discussions •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills •Marginally acceptable performance in assignments
F	•Demonstrate a poor understanding of the concepts involved •Unable or unwilling to handle questions •Minimal or no participation in discussions •Present arguments poorly •Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills •Poorly performance in assignments

Assessment Rubrics for Participation

A+, A, A-	•High participation in discussions •Always attend in-class discussions •Demonstrate a strong understanding of all relevant knowledge •Handling questions professionally •Present arguments that have an element of originality •Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	•Good participation in discussions •Often attend in-class discussions •Demonstrate a good understanding of all relevant knowledge •Handling questions in a logical way •Present arguments that go beyond the lecture and textbook •Respect others and follow the class rules (no chatting and do not use cell phone)
C+, C, C-	•Some participation in discussions •Sometimes attend in-class discussions •Demonstrate a basic understanding of the concepts involved •Fairly address questions as set •Present arguments in a well-structure manner •Respect others and follow the class rules (no chatting and do not use cell phone)

D+, D	•Minimal or no participation in discussions •Rarely attend in-class discussions •Demonstrate a minimum understanding of the concepts involved •Barely address questions as set •Present arguments in a marginally acceptable manner •Respect others and follow the class rules (no chatting and do not use cell phone)
F	•Minimal or no participation in discussions •Almost never attend in-class discussions •Demonstrate a poor understanding of the concepts involved •Unable or unwilling to handle questions •Present arguments poorly •Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments and the Exam	
A+, A, A-	•Demonstrate a strong understanding of all relevant knowledge •Present arguments that have an element of originality •Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	•Demonstrate a good understanding of all relevant knowledge •Present arguments that go beyond the lecture and textbook •Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	•Demonstrate a basic understanding of the concepts involved •Present arguments in a well-structure manner •Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	•Demonstrate a minimum understanding of the concepts involved •Present arguments in a marginally acceptable manner •Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	•Demonstrate a poor understanding of the concepts involved •Present arguments poorly •Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

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REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

To be announced on Moodle.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

- ○ Online response via Moodle site
- ○ Others: ___SFTL_____ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Assignments will be submitted in class before the lectures. No late assignment will be accepted.
Lecture notes and self-learning materials will be uploaded on Moodle.
Good questions and discussions on Moodle will be awarded participation grades.